1. General Information

1.1. Company Presentation

CompanyX was founded in 1922 by Joachim CompanyX in X-Stadt.

1.2. Mission and Objectives

CompanyX is seeking a partner for the planning and commissioning of an automated guided vehicle (AGV) system at the newly built logistics site in X-Str. 4 12345 X-Stadt. This will be a pilot project for the initial introduction of AGVs in CompanyX's contract logistics. The tender focuses on finding a partner for this project and other potential use cases in Germany. CompanyX also wants to learn about the providers' experiences at the European level to identify potential partners for future tenders in Europe.

CompanyX acts as the principal for the requirements described in this document. The tender covers project planning, implementation, production, delivery, assembly, commissioning, testing, and operational handover, including training, for a system that includes:

- Automated Guided Vehicles (AGVs)
- Operationally relevant peripherals (location stations, etc.)
- Control/management software for AGVs

The objective is for the supplier and contractor (Contractor) to take these requirements, translate them into an implementation concept, and execute it after the order is placed. The Contractor is given freedom to design the project, provided they adhere to the Client's basic requirements. Alternative solutions, especially those that reduce costs, must be specified by the Contractor in the offer, along with the potential financial effects.

1.3. Privacy Policy

The Contractor agrees to maintain secrecy about all operational information that is known or becomes known before the contract. This also applies to representatives and subcontractors hired by the Contractor.

1.4. Location

The logistics site is located in X-Str. 4 12345 X-Stadt. The warehouse area consists of six hall units with a total area of approximately 58,000 m².

1.5. Operations

Regular operation is scheduled from Monday to Friday, from 6:00 a.m. to 11:30 p.m. (two-shift operation). Due to seasonal trends, it may be necessary to operate outside these hours.

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2. Quoting

2.1. Bid Preparation

The preparation of the offer is free of charge and non-binding for the Client. No legal claim to the placement of the order or reimbursement of expenses for processing the offer can be derived from it. The bid must be submitted in full by the deadline. The offer price must be submitted according to the price sheet and must include all planning costs, wage and ancillary wage costs, other personnel costs, travel and hotel costs, expenses, set-up costs, assembly costs, tool and material costs, equipment and machine costs, commissions, and insurance. The unit prices quoted will be the basis for calculating any additional, increased, or decreased volume.

2.2. Applicable Documents

In addition to the offer, the Contractor must attach the following documents:

- Offer presentation
- Marked routes to be cleared for the AGV
- Completed price sheet in .xlsx format
- References of the Contractor
- Technical description of the AGVs and integrated components
- Suggestion for a Maintenance Offer
- Suggestion for a Service Offer
- Proposal for spare parts
- Detailed framework schedule with exemplary project progress
- Contact person of the Contractor

2.3. Alternative Offers and Proposed Changes

The submission of alternative offers or proposed changes is possible but will only be accepted as a supplement to the tendered scope of services. The full scope of the tendered services must be offered in any case. Alternative offers must be clearly marked as such and will only become part of the contract if they are included in an order letter after a corresponding agreement.

2.4. Awarding

The Client reserves the right not to commission individual parts of the scope of services and to award partial services separately from the total scope. A separate award does not entitle the Contractor to withdraw their offer or derive other claims from it.

2.5. Question Submission

Questions about the call for applications should be sent by e-mail only to the provided contacts using the questionnaire (Appendix 5). Fundamental information will be forwarded to all providers.

2.6. Other Conditions

The Contractor is obliged to comply with all relevant laws, decrees, regulations, technical regulations, and guidelines of the Federal Government and the State. The Contractor must check the validity of the tendered services against legal regulations.

2.7. Requirements for Subcontractors

A subcontractor list with service assignments must be enclosed with the tender.

2.8. Project Delivery Requirements

2.8.1. Project Organization: The Contractor must establish a project organization for the duration of the project, including at least a project manager and an assembly manager, each with a qualified deputy.

2.8.2. Meetings: A cyclical meeting of the core project team will take place, which the Contractor must attend. Additional meetings may also be scheduled.

2.8.3. Language: At all times during installation, the Contractor must have at least one person present who can communicate in German and is authorized to give instructions to other employees on site.

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3. Contact

3.1. Technical Issues

The contact persons for technical questions are:

3.2. Operations

The contact person for operations is the branch management at the X-Str. 4 12345 X-Stadt site, which will be named and announced at a later date.

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4. Scheduling

The planned schedule for the remainder of this tender is as follows:

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5. Price Sheet and Price Structure

The offer must be completed according to Appendix 1_Preisblatt Automated Guided Vehicles for both concept variants (see Chapter 6.). All options must be offered in full. The price sheet also includes cost items for the additional rental of vehicles for seasonal peaks and a cost indication for future projects at other CompanyX locations. Changes to the price sheet are not permitted.

The prices must include costs for the following services:

- Freight costs for all components
- Project management and contact person
- Travel expenses for contractors & subcontractors
- Mechanical and electrical assembly and commissioning
- Construction site security and equipment
- Implementation of interfaces to the WMS SAP
- Documentation
- Spare parts list
- Training
- Functional, performance, and availability tests
- Go-Live accompaniment
- Creation of an emergency outage concept

Missing items from this list are the Contractor's responsibility and expense. The prices should be broken down by individual components and services to allow for future fleet expansions. Maintenance and repair costs should be stated separately, on a per-year/per-month basis, with a proposed concept for maintenance and spare parts stocking.

6. Service Description Technology

CompanyX is pursuing two concept variants:

1. **Concept variant 1:** Exclusively ground transport
2. **Concept variant 2:** Connection of cantilever transfer stations to rack heads

The basic process is the same for both variants. The tender's requirements are functional, leaving the technical implementation to the bidder.

The two processes for pallet transport are:

1. Replenishment pallet from a narrow aisle to order picking
2. Outgoing goods pallet from a narrow aisle to shipping lines

6.1. Function and Process Description

Both processes have the same source transfer points, marked in red on the layout drawings. The pallets are single-type, foil-wrapped pallets. AGVs will operate on the main track (yellow), which is 4m wide and will have occasional forklift and intersection traffic.

6.1.1. Source Transfer Stations for Both Processes

Pallets are retrieved semi-automatically by an induction-guided narrow-aisle forklift and placed at the rear transfer points on the rack for AGV pickup. The transfer stations are managed as buffer locations in the WMS. When a pallet is placed, a pickup order for the AGV is automatically created via the WMS interface. The AGV should pick up the pallets according to priority and the FiFo principle.

Pallet Pickup: The AGV must be able to pick up pallets either lengthwise from the roadway or from the aisle on the short side with floor-level entry openings. CompanyX asks for a recommendation on which variant is best for profitability and process reliability.

Pallet Detection: The AGV should use environmental scanners to identify the exact position of the pallet's entry openings and align itself accordingly. It should also be able to detect twisted pallets and identify the entry opening to avoid generating an error message.

Cantilever Transfer Stations: In Concept Variant 2, cantilever transfer stations at heights of 2700 mm and 5400 mm are to be integrated. These stations are 1.25m wide and 1.6m long.

Quantity Data:

- **Concept Variant 1 (Ground Transport):** Pallets are picked up at floor level only. The number of floor buffer spaces would be increased to two per shelf row, totaling 196 source transfer locations.

	Narrow aisle hall 1	Narrow-aisle hall 5	Narrow-aisle hall 6
Number of individual shelf rows	30	30	38
Number of floor spaces	2	2	2
Total source transfer locations	60	60	76
Total source transfer locations			**196**

- **Concept Variant 2 (Cantilever Stations):** Cantilever transfer spaces are also integrated, requiring vehicles with a lift. This would provide one buffer space per level, for a total of 294 source transfer locations.

	Narrow aisle hall 1	Narrow-aisle hall 5	Narrow-aisle hall 6
Number of individual shelf rows	30	30	38
Number of floor spaces	1	1	1
Number of cantilever workstations	2	2	2
Total source transfer locations	90	90	114
Total source transfer locations			**294**

6.2. Sink Transfer Stations for Replenishment Pallets in Picking

Replenishment pallets are placed on transfer points in the picking halls (Hall 2 and Hall 3). The AGV should place the pallet at the nearest free transfer location to the target storage bin. The status of the buffer station must be transferred to the WMS as "occupied" with the pallet number. Hall 2 has only floor transfer stations, while Hall 3 has additional cantilever transfer stations.

Quantity Data:

- **Concept Variant 1 (Ground Transport):** Pallets are handed over at floor level only. The number of floor buffer spaces would be increased to two per shelf row, totaling 76 sink transfer places.

	Picking Hall 2	Picking Hall 3
Number of individual shelf rows	18	20
Number of floor spaces	2	2
Total sink transfer points	36	40
Total sink transfer points		**76**

- **Concept Variant 2 (Cantilever Stations):** Cantilever transfer spaces are integrated, requiring vehicles with a lift. This would provide a total of 96 sink transfer places.

	Picking Hall 2	Picking Hall 3
Number of individual shelf rows	18	20
Number of floor spaces	2	1
Number of cantilever workstations	0	2
Total sink transfer points	36	60
Total sink transfer points		**96**

6.3. Sink Transfer Stations for Outgoing Goods Pallets on Shipping Lines

Outgoing goods pallets are placed on a block transfer area in Hall 2, where they are sorted by a forklift driver onto various shipping lines. Separate transfer areas will be set up for two delivery types: "direct delivery" and "early purchase". These transfer areas are identical for both concept variants, with a total of 30 transfer spaces.

6.4. Interface to the WMS (LFS 400 from EPG)

The LFS400 from Erhardt + Partner is the leading WMS, containing all necessary information for orders and handling units. Communication is required between the LFS and the AGV's control software. The exact interface definition will be detailed after the contract is awarded.

6.5. Pallet Specifications

The pallets are dimensionally stable, foil-wrapped wooden pallets with cardboarded goods. Most pallet weights are less than 500 kg, with a maximum of 1,000 kg. The AGV concept can be limited to loading equipment categories 111-224 (120×80 cm and 120×100 cm), which account for the majority of movements.

6.6. System Performance

The system must be able to perform the specified pallet transports within 16 hours, including time losses due to IT waiting times. The quantity data provided (Figure 14) only includes pallets in the 111-224 loading equipment categories.

	Source	Depression	Distance single [m]	Distance double [m]	Number of pallets/year	Number of pallets/day
Replenishment pallet in order	U06	U03	15	30	8,444	34
	U05	U03	160	320	7,155	28
	U01	U03	150	300	20,423	81
Outgoing goods pallet in loading	U06	U02	60	120	14,015	56
	U05	U02	260	520	1,370	5
	U01	U02	50	100	37,435	149

6.7. Availability

The technical system availability of the entire system should be at least 98.5%.

7. Technical Requirements

7.1. Floor Condition

The concrete floor slab is designed for logistics operations with flatness according to DIN 18202. A gradient of approximately 1.5% must be taken into account near the loading gates. The contractor must verify all dimensions on-site after the contract is awarded.

7.2. Network Infrastructure

The AGV should preferably communicate via 5 GHz radio.

7.3. Safety Devices & Escape Routes

The scope of delivery includes all necessary safety equipment according to ASR, BGV, and UVV regulations, such as protective grilles and fall protection devices. The AGVs must not block fire protection doors or other escape routes in the event of a fire.

7.4. Emergency Stop Functions

Emergency stop switches must be provided in the required number and located in an accessible manner on the AGVs. Emergency stop functions must comply with applicable regulations and rules.

7.5. Power Supply

The Contractor is asked to specify the total electricity capacity and other energy inflows required for the system. The main supply cables will be laid to the main distribution board, but the contractor is responsible for the connection and the entire fixed installation of electrical components.

7.6. Control Software & Remote Maintenance

Control software with a visualized interface is required to show the AGVs' driving orders and positions. All messages must be displayed in German plain text. A remote diagnosis option must be included for faults that cannot be rectified on site.

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8. Requirements for Commissioning

8.1. Trial Operation/Test Phase

Before final commissioning, a trial operation of at least four weeks is to be carried out as a test phase. The trial can be done with one device and reduced complexity to verify assumptions and demonstrate cooperation between employees and the AGV. The trial operation must be priced separately as an "all-in rate".

8.2. Acceptance and Performance Test

A preliminary acceptance will be carried out based on functional and performance tests. The Contractor will prepare the test program and record the tests. The transfer of risk begins after successful preliminary acceptance.

8.3. Trial Operation (Normal Operation), Availability and Final Acceptance

Following preliminary acceptance, trial/normal operation will take place under the Contractor's supervision to prove the system's characteristics, achieve a stable process, and train employees. Final acceptance will occur after successful trial operation. An availability test will be conducted on two consecutive days to check the reliability of all components.

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9. Training and Instruction

The Contractor will train designated persons in the operation and simple maintenance of the systems. A separate group (team leaders, technicians) will be specially trained in fault elimination and maintenance. The training topics should cover:

- Function and structure of the AGV system (incl. software)
- Operation of the AGVs (incl. software)
- Explanation of mechanical functions (maintenance)
- Behavior in error situations and rectification

The training will take place on site before live operation.

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10. Documentation

The Contractor must provide all documentation to allow the Client to operate the plant properly and safely, identify and resolve malfunctions, and maintain the system. All documents must be in German and provided in printed and electronic form. The documentation should include:

- Detailed drawings of vehicles and charging stations
- Electrical connected loads
- Operating instructions
- Repair/replacement instructions
- Safety information
- Test Protocols
- EU Declaration of Conformity

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11. Services

The offer must include a proposal for a maintenance concept and a necessary wear and spare parts package.

11.1. Spare & Wear Parts

Spare parts should be classified as: Risk parts, Recommended spare parts, and Consumables. Spare parts not stocked by CompanyX must be delivered within a maximum of 24 hours. The Contractor guarantees the availability of spare parts for at least 10 years after final acceptance.

11.2. Maintenance & Repair

The tender must include a draft maintenance contract covering inspections, maintenance, and checks. Maintenance work must be scheduled to avoid disrupting ongoing operations.

11.3. Remote Maintenance

The Contractor must ensure remote diagnosis and maintenance are possible for malfunctions in the control technology.

11.4. On-call Duty and Troubleshooting

The Contractor must offer on-call duty from 6:00 a.m. to 10:00 p.m., Monday to Friday, and for critical cases leading to a standstill, even outside these times. A telephone hotline for system handling, troubleshooting, and administration is also required. The Contractor's reaction time is one hour on weekdays and six hours on weekends for a technician to arrive on site.

11.5. Claims for Defects and Liability

Liability for property damage is limited to €2.5 million per claim and €5 million per year. Unlimited liability applies to pure financial losses.

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12. Annexes to the Specifications

- Appendix 01: Price sheet for concept variant 1 (ground transport)
- Appendix 02: Price sheet for concept variant 2 (cantilever parking spaces)
- Appendix 03: Quantities & Distances
- Appendix 04: Layout plan
- Appendix 05: Provider Questionnaire
- Appendix 06: Questionnaire

original document within the text.